



SOIL EROSION AND CONSERVATION — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

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4. Soil Erosion Process Order (Splash→Sheet→Rill→Gully)
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| 1. Critical Soil Erosion Areas — Shivalik Foothills, Chambal

- India ke max soil-erosion-affected areas: **Chambal-Yamuna ki barren land**, **Western Himalaya/Shivalik foothills**, **Chotanagpur Plateau**, **Tapti-Sabarmati ke beech (Malwa Plateau)**, **Maharashtra ka black soil area**, **Haryana-Rajasthan-Gujarat ke arid regions** — **Shivalik foothills mein soil erosion sabse critical hai.** [Q1]

| 2. Chambal Valley — MP's Erosion Hotspot (Morena)

- India mein **Chambal Valley soil erosion se sabse zyada affected** hai — Madhya Pradesh mein specifically **Morena district** (Bhind, Gwalior, Shivpuri bhi) affected hai — 'Chambal ravines' banti hain. [Q2, Q3]

| 3. Gully Erosion — Chambal Ravines Cause

- Chambal Ravines (bad-lands) **Gully erosion** ki wajah se bante hain — Splash, Sheet, Rill erosion se nahi. [Q4]

| 4. Soil Erosion Process Order (Splash→Sheet→Rill→Gully)

- Soil erosion process ka sahi order: **Splash erosion (raindrop impact)** → **Sheet erosion (thin uniform layer removal)** → **Rill erosion (small straight channels)** → **Gully erosion (rills se badi channels)** — aage **Stream channel erosion** bhi ho sakti hai. [Q5]

| 5. Sorghum — Most Erosion-Prone Crop

- WWF analysis ke mutabik **Sorghum (jwar) ugane par farmland sabse tezi se erode hoti hai** — **Clover sabse kam erosion karti hai** (Potato, Wheat in beech mein hain). [Q6]

| 6. Soil Fertility Methods — Crop Rotation vs Seed Modification

- Soil fertility improve karne wale sahi methods: **Crop rotation, Mixed cultivation, Multiple cropping** — **Seed modification soil fertility ko affect NAHI karta (sirf crop production ko affect karta hai)**. Crop rotation ka main purpose **soil fertility badhaana** hai (pest control mein bhi madad karta hai). [Q7, Q8]

7. Soil Conservation Practices (Terracing, Windbreaks)

- India mein appropriate soil conservation methods: **Crop rotation, Terracing, Windbreaks** — **Sand fences appropriate NAHI hai India ke context mein**. [Q9]

8. Soil Erosion Causes — Deforestation (Not Tropical Climate)

- Soil erosion ka problem **Deforestation se associated hai** — **Terrace cultivation ek conservation method hai (erosion ka cause NAHI), aur Tropical climate bhi erosion ke liye responsible NAHI hai**. [Q10]

⚠ EXAM TRAP — Terrace cultivation — erosion ROKNE ka method hai, erosion ka CAUSE nahi hai

9. Afforestation — Primary Erosion-Check Method

- Soil erosion **Afforestation (vanikaran) se check ki jaa sakti hai** — **Excess grazing, plant removal, bird-count badhaana ise ROKte NAHI hain, balki contour tillage, shifting-cultivation ban bhi effective methods hain**. [Q11, Q12]

10. Water Conservation Practices — Zero Tillage & Gypsum

- Agriculture mein water conservation ke sahi practices: **Reduced/zero tillage (soil moisture conserve karta hai), Gypsum application before irrigation (salt-clay-pan dissolve karta hai, percolation improve karta hai), crop residue field mein chhodna (evaporation kam karta hai)** — **teeno sahi hain**. [Q13]